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ARTICLE

# Differential recovery of δ¹³C in multiple tissues of white sucker across age classes after the closure of a pulp mill

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**Abstract:** Techniques to document recovery after the closure of pulp mills that discharge enriching effluents are not well established, but $\delta^{13}C$ may be a useful tool. In the 1990s, the muscle tissue of white sucker (*Catostomus commersonii*) collected downstream of two pulp and paper mills discharging into separate streams (Mattagami and Kapuskasing rivers) was enriched in $^{13}C$ compared with upstream fish, suggesting uptake of pulp-derived C. The Mattagami River mill was closed in 2006, and analysis of muscle and gonad for $\delta^{13}C$ was performed in 2011. As expected, fish captured in 2011 downstream of the operational Kapuskasing mill still showed the influence of the pulp-derived C in muscle and gonad tissue. After the closure of the Mattagami River mill, muscle tissue of white sucker was still enriched in $^{13}C$ compared with upstream fish, while gonad tissue was not. The patterns observed in the Mattagami River were, however, related to age; the oldest fish showed enrichment of $\delta^{13}C$ in both muscle and gonad tissue, suggesting the residual occurrence of pulp-derived C. This study suggests that measurements of stable isotopes in fish across a broad age range may indicate ecosystem improvements. These techniques may also be useful where no data prior to the upgrade or closure are available for comparison.

**Résumé :** S'il n'y a pas de technique bien établie pour documenter le rétablissement après la fermeture d'usines de pâte qui relâchent des effluents enrichissants, le $\delta^{13}C$ pourrait s'avérer utile. Dans les années 1990, des tissus musculaires de meuniers noirs (*Catostomus commersonii*) prélevés en aval de deux usines de pâte et papier relâchant des effluents dans des ruisseaux différentes (Mattagami et Kapuskasing) étaient enrichis en $^{13}C$ par rapport à ceux de poissons en amont des usines, semblant indiquer une absorption de C provenant de la pâte. L'usine sur la rivière Mattagami a fermé ses portes en 2006, et les analyses ont été répétées en 2011. Comme prévu, leurs muscles et gonades de poissons capturés en aval de l'usine toujours en exploitation sur la rivière Kapuskasing présentaient encore une influence du C provenant de la pâte. Après la fermeture de l'usine de la rivière Mattagami, les tissus musculaires de meuniers noirs étaient toujours enrichis en $^{13}C$ par rapport à ceux de poissons en aval, alors que les tissus gonadiques ne l'étaient pas. La distribution du $^{13}C$ observée dans la rivière Mattagami était toutefois reliée à l'âge, les muscles et les gonades de poissons plus vieux présentant un enrichissement du $\delta^{13}C$, suggérant la présence résiduelle de C provenant de la pâte. L'étude donne à penser que la mesure d'isotopes stables dans des poissons d'âges très variés pourrait permettre de documenter l'amélioration d'écosystèmes. Ces techniques pourraient également être utiles quand aucune donnée antérieure à la mise à niveau ou la fermeture n'est disponible pour des fins de comparaison. [Traduit par la Rédaction]

## Introduction

During the operation of pulp mills, organic enrichment can occur in the environments that receive the effluent (Chambers et al. 2000). The effects of this enrichment can be documented by studies on the health of fish or invertebrates (Lowell et al. 2003), and the exposure can be measured by the incorporation of nutrients like carbon (C) or nitrogen (N) into the tissues of organisms (Wayland and Hobson 2001). Typically, a terrestrial $\delta^{13}C$ signature (near −28‰; France 1995) is found in the tissues of aquatic organisms exposed to pulp mill effluents (Wassenar and Culp 1996) and $\delta^{15}N$ is near 0‰ (Wayland and Hobson 2001). Although patterns of exposure are often observed in organisms captured downstream of operating pulp mills, what occurs after the facility ceases discharging effluent is not known.

After the closure of an industrial facility discharging wastewater with measurable impacts, recovery is expected in the receiving environment. Since $\delta^{13}C$ is often useful in documenting the exposure of effluents on receiving environments during the operation of a pulp mill, it may be an advantageous tool to document the abating influence of an effluent after closure of the mill. Changes in the ratio of stable isotopes would be expected if the effluent was an important source of nutrients during the operation of the mill. Documenting shifts in $\delta^{13}C$ would then be important in determining if an ecosystem has recovered from previous exposure to an enriching effluent.

Measurements of isotopic shifts in soft tissues of living organisms have been used as monitoring tools to trace the specific influence of effluents from operating facilities on ecosystems (i.e., Tucker et al. 1999). Most studies examining this phenomenon have observed changes in $\delta^{15}N$ after closure or treatment upgrades at sewage plants; after upgrading to tertiary treatment, the altered $\delta^{15}N$ discharged by a sewage plant was detectable as a corresponding shift in the tissue of short-lived marine algae with fast turnover (Rogers 2003; Savage and Elmgren 2004) and less so in longer-lived organisms with slower turnover, such as limpets and mussels (Rogers 2003). These data suggest the persistence of signals linked with previous effluents in the tissues of organisms because of different rates of turnover that may be related to body size (Weidel et al. 2011). If turnover rates slow with age and in-

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creasing body size, then soft tissues measured in fish along an age (and size) gradient may be useful to detect residual influence of an effluent after the closure of an industrial facility. These patterns may be detectable downstream of pulp mills by examining changes in $\delta^{13}C$ over time.

Sampling multiple soft tissues within a single individual may also provide important information on ecosystem changes after the closure of a pulp mill. In diet switching studies, organisms such as fish can show fast equilibration of their tissues with the new diet (Hesslein et al. 1993), but the tissues sampled within these studies may also have an influence on the detected turnover rates (Perga and Gerdeaux 2005). $\delta^{13}C$ of fish muscle should respond more slowly than more metabolically active tissues such as gonad, but these have not been measured concurrently in large-bodied fish in a study of ecosystem recovery.

Hard tissues have also been used to document changes in ratios of stable isotopes over time and may be useful in recovery studies. In fish, hard tissues with slow turnover, such as scales (Trueman et al. 2012) and otoliths (Grønkjær et al. 2013), have been used as archives of stable isotopes, which can be measured in individual annuli several years later (Trueman et al. 2012). Few studies, however, have used hard tissues to examine historical patterns linked to a specific anthropogenic activity (Farwell 2000), although some have examined large-scale phenomena (Trueman et al. 2012). An advantage of using hard tissues, such as opercula or other aging structures, is the access to historical information contained within each annulus.

A pulp mill discharging to the Mattagami River had varying impacts during the 1990s, including underlying effects of nutrient enrichment (Munkittrick et al. 2000; Stantec 2007). It was closed in 2006. The general purpose of this paper was to evaluate $\delta^{13}C$ as a tool for documenting recovery of ecosystems after exposure to pulp mill effluents. Work was initiated in 2011 to determine if residual signals in $\delta^{13}C$ from the pulp mill were still present in multiple tissues of fish along a broad age gradient. The specific purposes of this study are twofold: (*i*) to compare historical (1990s) and recent (2011) $\delta^{13}C$ of muscle in white sucker (*Catostomus commersonii*) from above and below a closed pulp mill on the Mattagami River to determine if exposure to effluents from the pulp mill resulted in distinct isotope ratios in these fish; (*ii*) to determine changes in the $\delta^{13}C$ of gonad, muscle, and bone in multiple age classes of fish captured downstream of the closed mill in 2011. We predicted that fish captured downstream of the pulp mill would have distinct $\delta^{13}C$ of tissues indicative of the influence of the effluent relative to upstream, with the exception of the recent (2011) sampling period following the closure of the Mattagami mill. We predicted that older fish captured downstream of the closed mill in 2011 would show different $\delta^{13}C$ compared with fish captured at the upstream site in slow-turnover muscle tissue, but similar $\delta^{13}C$ in more recently synthesized gonadal and opercular tissue.

## Methods

The study area is in the Moose River Basin in northeastern Ontario, Canada (Fig. 1). The Moose River drains 90 643 km² and flows north to James Bay. A tributary of the Moose River, the Mattagami River, is sixth order, 491 km long, and drains an area of 35 612 km² (Brousseau and Goodchild 1989). The Kapuskasing River is a tributary of the Mattagami River, is fifth order, 324 km long, and has a drainage area of 8633 km² (Brousseau and Goodchild 1989). Hydroelectric developments are present at the town of Kapuskasing on the Kapuskasing River and at Smooth Rock Falls (SRF) on the Mattagami River adjacent to pulp mill sites, but upstream of the outfalls. The dams at SRF and Kapuskasing are run-of-the-river facilities and do not have substantial headponds (maximum depth ≤12 m).

Two pulp mills, one recently closed (Mattagami) and one currently operational (Kapuskasing), were studied. The closed mill at SRF was built in 1917 as an unbleached kraft mill. A hydroelectric dam was also built during this period to fulfil the power and water requirements of the mill. The mill was converted to a bleached kraft mill in the 1930s (Stantec 2007). At its peak, the mill produced 465 air-dried metric tons (t) of pulp and discharged 70 000 m³ of effluent per day. The secondary treated effluent was discharged into the Mattagami River 0.15 km below the hydroelectric dam via two submerged multiport diffusers. The mill was idled in July 2006, closed permanently in November of that year, and dismantled in 2008. Sampling was also done at a second pulp mill that was operating during both study periods. The thermomechanical pulp mill at Kapuskasing began operations in 1909. The mill produces ~1050 t per day of newsprint and 225 t per day of recycled pulp. Secondary treatment of effluent via activated sludge began in April 1995. Mill effluent (~70 000 $m^3 \cdot day^{-1}$) is discharged into the Kapuskasing River 0.1 km downstream of the dam.

Water chemistry was measured intermittently in both the Mattagami and Kapuskasing rivers upstream and downstream of the pulp mill outfalls (Table 1). There were transiently higher nutrients at the downstream sites in both rivers. The largest differences were observed in particulate N in the Mattagami River and total Kjeldahl N in both the Mattagami and Kapuskasing rivers at the downstream (MAT-DS and KAP-TB) compared with the upstream (MAT-US and KAP-WF) reference sites. Values for these nutrients are indicative of the oligo- to mesotrophic nature of boreal rivers (Smith et al. 1999).

The species chosen for this work was the white sucker. It is abundant and is widely distributed in the watershed. White sucker also feeds on benthic invertebrates, which increases the likelihood of encountering contaminants found locally in the sediment downstream of a diffuser and with increasing exposure. This species has been used in a number of fish population studies, including the extensive work at Jackfish Bay (Munkittrick et al. 1991), a survey of 10 pulp mills (Munkittrick et al. 1994), and the previous work at the Mattagami mill (Munkittrick et al. 2000; Stantec 2007). The reference and exposed sites in the Mattagami and Kapuskasing rivers are separated by run-of-the-river dams, which prevent fish from migrating upstream.

### Study periods

#### *1990s fish sampling*

During the 1990s white sucker were captured using gill nets (8.8 and 10.2 cm mesh size). Sampling in the Mattagami River occurred in the fall of 1994 (15–27 September), 1996 (11–24 September), and 1997 (12–26 September). The reference site in the Mattagami River, MAT-US, is located 3–5 km above the dam in a reservoir (Fig. 1). The exposed site, MAT-DS, is downstream of the dam and spans a distance of 1–3 km below the mill's diffuser. The Kapuskasing River was sampled in the fall of 1994 and 1996. This sampling included reference (KAP-WF) and exposure (KAP-TB) sites. KAP-WF was sampled in 1994 and 1996 and is above a set of rapids (Woman Falls) approximately 40 km upstream from the effluent discharge. KAP-TB was sampled in 1994 and 1996 and is located <1 km from the Kapuskasing mill's diffuser downstream of the dam.

#### *2011 fish sampling*

In 2011 white sucker were collected in September (6–15) at the MAT-US and MAT-DS sites in the Mattagami River and at the KAP-TB and KAP-WF sites in the Kapuskasing River. To capture more young fish indicative of recent conditions, 7.6 cm mesh was added to each gillnet gang of 8.8 and 10.2 cm mesh nets used in the 1990s. Fish collections in 2011 were approved by the Animal Care Committee at the University of New Brunswick, Saint John (UNBSJ ACC, 2011-2D-01).

**Fig. 1.** Study area in the Moose Basin in northern Ontario, Canada; inset shows the location of sites sampled for white sucker during the 1990s and 2011. KAP-WF = Kapuskasing River above Woman Falls (reference site); KAP-TB = Kapuskasing below Two Bridges motel (exposed); MAT-US = Mattagami River upstream of pulp mill (reference site); MAT-DS = Mattagami River below pulp mill (exposed).

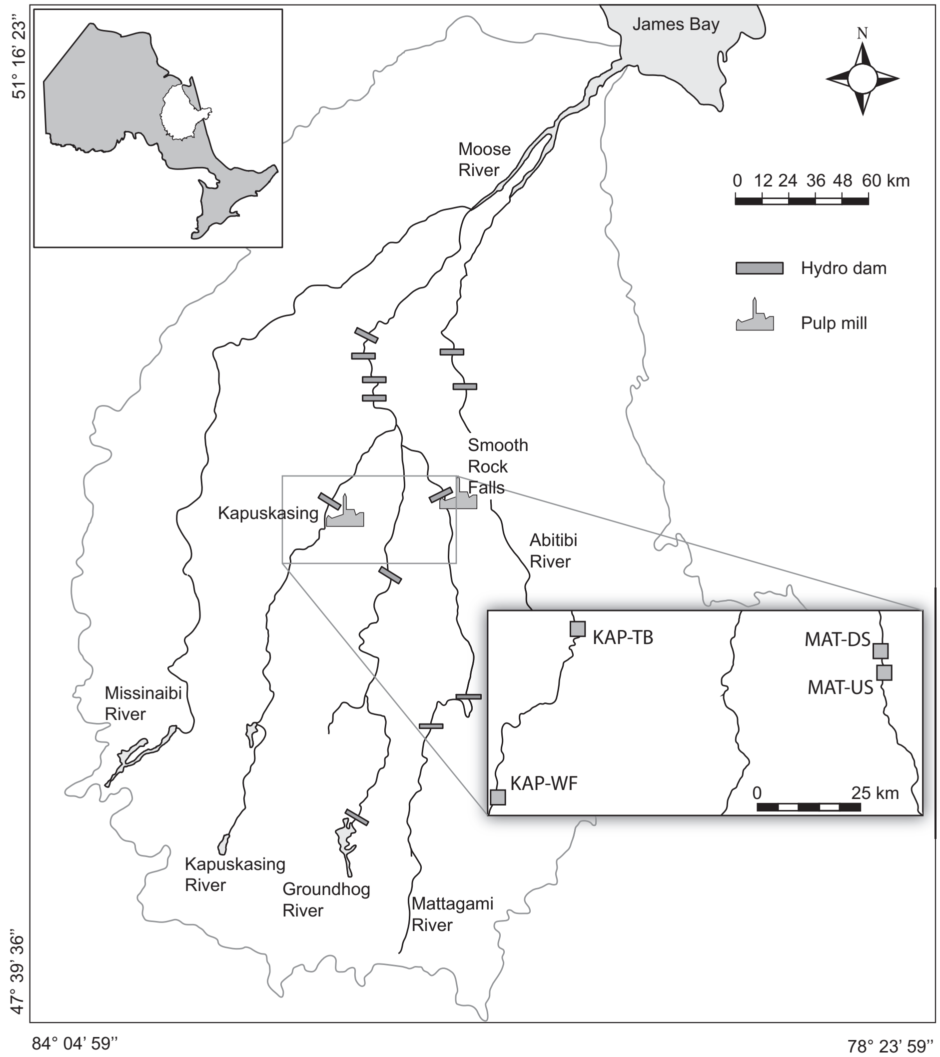



**Table 1.** Water chemistry of selected variables measured in the Mattagami (MAT-US and MAT-DS) and Kapuskasing (KAP-WF and KAP-TB) rivers in 1994–1996 and 2011.

| Site | Year | Dissolved P (mg·L⁻¹) | Particulate P (mg·L⁻¹) | Dissolved N (mg·L⁻¹) | Particulate N (mg·L⁻¹) | TKN* (mg·L⁻¹) |
|---|---|---|---|---|---|---|
| MAT-US | 1994 | 16.3 | 17.3 | 406 | 53 | — |
| | 1995 | 11.0 | 18.3 | — | 55 | 539 |
| | 1996 | 17.9 | 17.7 | — | 74 | 1506 |
| | 2011 | <5.0 | — | — | — | 379 |
| MAT-DS | 1994 | 15.8 | 16.2 | 439 | 69 | — |
| | 1995 | 10.3 | 18.4 | — | 68 | 652 |
| | 1996 | 31.1 | 22.3 | — | 233 | 1508 |
| | 2011 | <5.0 | — | — | — | 363 |
| KAP-WF | 1994 | 10.1 | 9.2 | 480 | 100 | — |
| | 1995 | 9.3 | 12.1 | — | 68 | 494 |
| | 1996 | 8.8 | 14.9 | — | 37 | 907 |
| | 2011 | <5.0 | — | — | — | 413 |
| KAP-TB | 1994 | 16.2 | 14.4 | 440 | 67 | — |
| | 1995 | 14.6 | 15.1 | — | 74 | 661 |
| | 1996 | 8.8 | 14.9 | — | 54 | 1236 |
| | 2011 | 24.4 | — | — | — | 566 |

**Note:** Data from 1990s adapted from Munkittrick et al. 2000.
*Total Kjeldahl nitrogen.

**Table 2.** Mean $\delta^{13}C$ ± SE (*n*) in muscle tissue of white sucker captured in the Mattagami (MAT) and Kapuskasing (KAP) rivers in the 1990s.

| | | Year | | | |
|---|---|---|---|---|---|
| River | Site | 1994 | 1995 | 1996 | 1997 |
| MAT | MAT-US | −37.05±0.77 (10) | −34.78±0.63 (10) | −34.61±0.99 (4) | −34.85±0.93 (10) |
| | MAT-DS | −27.20±0.54 (4) | −27.41±0.45 (4) | −27.62±0.59 (4) | −27.53±0.47 (10) |
| KAP | KAP-WF | −31.98±0.22 (4) | | −30.06±0.54 (4) | |
| | KAP-TB | −25.30±0.20 (4) | | −24.78±0.46 (4) | |

**Note:** No statistics were performed on these data. US, upstream; DS, downstream; WF, Woman Falls; TB, Two Bridges.

## Isotope samples

### *Muscle and gonad tissue*

In the 1990s samples of dorsal white muscle of white sucker (5–10 g) were collected from four to ten individuals per site, wrapped in precombusted foil (400 °C), placed in Whirlpac bags, and stored on ice in the field. All samples were stored at −20 °C prior to preparation for isotope analyses shortly after the fish were collected. The samples were freeze-dried, ground into a powder, and analyzed for carbon isotope ratio ($^{13}C/^{12}C$). In the 1990s the samples were analyzed at the University of Waterloo (Waterloo, Ontario) Environmental Isotope Laboratory.

In 2011, muscle and gonad samples were dissected, placed into scintillation vials, kept on ice for <8 h, and stored at −20 °C. The muscle and gonad samples were freeze-dried (Labconco Inc.) for 48 h. Samples of dorsal muscle from white sucker were not lipid-extracted because this procedure had no significant effect on the value of $\delta^{13}C$ ($p > 0.05$; Farwell 2000). Although there are obvious differences in the C:N ratio between sexes within a tissue (i.e., ranges of male and female gonad tissue at MAT-US are 3.15 to 3.76 and 4.03 to 5.23, respectively), these differences did not affect the outcome of our study. In 2011, the samples were submitted to the Stable Isotopes in Nature Lab (SINLab) at the University of New Brunswick (Fredericton). The mean difference of $\delta^{13}C$ in duplicates of gonad ($n = 14$) and muscle ($n = 11$) from fish collected in 2011 was 0.13‰ ($p = 0.096$; paired $t$ test) and 0.14‰ ($p = 0.414$; paired $t$ test), respectively. The mean difference of reference materials used to monitor the analytical precision was 0.06‰ for $\delta^{13}C$. Although $\delta^{15}N$ was measured in each sample, it was not used in the current analyses because it showed no discernible pattern among sites and times in the 1990s.

To determine the isotope signal from the pulp mill, effluent samples were also collected for stable isotope analysis. A sample was collected from the Kapuskasing mill in the fall of 1994. Two samples of effluent from the Mattagami mill were collected in the fall of 1996 and one sample was collected in the spring of 1995. Effluent was obtained (20 L) and filtered through precombusted (400 °C) GF/C filters. The filters were wrapped in precombusted foil and stored at −20 °C prior to preparation for stable isotope analysis. In the laboratory the material was removed from the filters, freeze-dried, and ground into a powder. These samples were also analyzed for ratios of stable C and N at the Environmental Isotope Laboratory at the University of Waterloo. Analytical precision (1 SD), based on repeated analyses of International Atomic Energy Agency and National Institute of Standards and Technology standards, was estimated as ±0.2‰ for $\delta^{13}C$.

### *Aging and opercular $\delta^{13}C$*

White sucker were aged using left opercula (Ruemper 1998). To determine if we could detect a change in the isotopic composition of bone deposited following mill closure, in addition to the muscle and gonad samples, we analysed right opercula from fish at the MAT-US and MAT-DS sites. Previous work suggested that temporal changes in $\delta^{13}C$ could not be easily traced using sequential annuli from individual fish (Farwell 2000). Rather than a sequential approach, we used the second annulus from 40 fish from the reference (MAT-US) and 44 fish from the exposed (MAT-DS) sites. The second annulus was selected because of its relative size, because white suckers begin to sexually mature at 3 years old, and because it is assumed to be free of maternal tissue. The bone sample was excised using a 4 mm diameter punch plier. Only $\delta^{13}C$ was used from the opercular bone since the bone had a low (<5%) N content. The mean difference among duplicates of the opercular bone was higher than the samples of muscle or gonad. From the nine duplicate samples of bone analyzed, the mean difference for $\delta^{13}C$ was 0.96‰. This error was considered acceptable to detect changes, since the difference in $\delta^{13}C$ in muscle tissue in fish from MAT-US and MAT-DS in the 1990s was approximately 7‰.

## Data analyses

The primary analysis in this study is the relationship of $\delta^{13}C$ in muscle tissue among periods within sites. To streamline the analysis of $\delta^{13}C$ among periods within sites, we first determined if sexes could be pooled. We found, using ANOVA, that there were no differences between sexes at any sites in either the 1990s or in 2011 ($p \geq 0.063$). For all further analyses, a pooled group of males and females was used. We also determined if $\delta^{13}C$ among years in the 1990s could be pooled. We found no significant differences between years at MAT-US ($p = 0.13$) or MAT-DS ($p = 0.97$), but sample sizes in some years were low (i.e., $n = 4$). The pooled value of $\delta^{13}C$ in the muscle of white sucker within sites was used for the analysis of changes among periods. The data from individual sites are presented for comparative purposes in Table 2.

A further analysis within this work was the comparison of gonad and muscle tissue in 2011. Because of differences in the rate of tissue turnover, we predicted that muscle, but not gonad, would show residual $\delta^{13}C$ from the effluent prior to closure. However, turnover rates of various tissues in large fish have not been clearly established (Weidel et al. 2011). To determine the specific relationships in $\delta^{13}C$ among sites (i.e., MAT-US, MAT-DS), tissues (muscle and gonad), and period (1990s and 2011), a tiered approach was required. This approach was used because gonad tissue was not sampled during the 1990s and the complete three-way ANOVA was not possible. Two two-way ANOVAs were used to assess the differences in values of $\delta^{13}C$ in fish captured in 2011 and in the muscle tissue of white sucker captured between periods. The first two-way ANOVA analyzed the effects of site and tissue on the $\delta^{13}C$ in white sucker captured in 2011. A second two-way ANOVA to determine the temporal change in the isotopic signature of C in muscle used site and period as factors.

The same approach used to analyze the $\delta^{13}C$ of white sucker captured in the Mattagami River was used in the Kapuskasing River. An initial ANOVA was used to determine if there were differences in $\delta^{13}C$ attributable to sex ($p \geq 0.051$; this $p$ value was found at KAP-TB: male $\delta^{13}C$ was −29.12‰ ± 0.39‰, and female $\delta^{13}C$ was −30.22‰ ± 0.36‰; this marginal significance is not considered biologically relevant). A second ANOVA was done on tissue and among fish collected in 2011. The role of individual year in the 1990s was also examined in the Kapuskasing River within sites in muscle tissue. Similar to the Mattagami River, there were no significant differences at KAP-TB ($p = 0.33$), but there was a statistical difference at KAP-WF ($p = 0.017$). Because the temporal trend was intact (i.e., $^{13}C$ enrichment at KAP-TB compared with KAP-WF in

1994 and 1996), the δ¹³C between sites was pooled for years for the temporal analysis of study periods.

To determine the temporal relationships in the δ¹³C of muscle tissue in white sucker, a two-way ANOVA was performed using site and period as factors. Sex information was not available for fish collected in the Kapuskasing River during the 1990s; thus, sex was excluded as a factor in this ANOVA.

We predicted that age of white sucker may affect the δ¹³C of fish tissues following the closure of the pulp mill because of changes in growth and turnover that accompany aging. A linear regression was used to determine the predictive relationship between age of fish and δ¹³C of tissues at each site. Analysis of covariance (ANCOVA) was used to determine differences in the relationships between age and δ¹³C between tissues at sites that showed significant regressions (i.e., MAT-DS and KAP-TB). Linear regression was also used to determine the change in the δ¹³C of opercula over time (δ¹³C of second annulus compared against year). In the analysis of opercular δ¹³C, we determined whether males and females could be pooled using ANOVA; there were no differences between the opercular δ¹³C of males and females at either MAT-US or MAT-DS ($p \geq 0.22$), and the sexes were pooled within sites for further analysis. ANCOVA was used to determine if there were differences in δ¹³C of the second annulus within each site with year of growth as the covariate.

All statistics were performed with Systat 13 and used a significance level ($\alpha$) of 0.05.

## Results

### Pulp mill effluent δ¹³C

Effluent was collected from both the Kapuskasing and Mattagami mills during the initial studies of the 1990s and indicated a range in δ¹³C of –24.5‰ to –26.8‰. In the fall of 1994, the Kapuskasing mill effluent had a δ¹³C of –26.8‰. In the effluent from the Mattagami mill, the δ¹³C was –26.2‰, –25.2‰, and –24.5‰ in the spring of 1995, the fall of 1994, and the fall of 1996, respectively.

### 1990s fish muscle δ¹³C

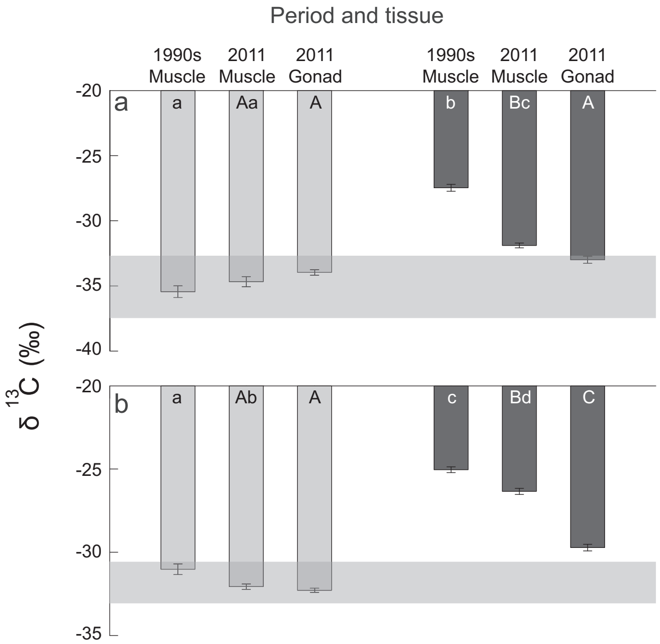
Elevated δ¹³C was found in the tissues of white sucker captured downstream of the Mattagami and Kapuskasing mills in all studies during the 1990s (Table 2), suggesting exposure to pulp mill effluent in all years. The mean value of the δ¹³C at the reference site in the Mattagami River was –35.45‰ ± 0.44‰ ($n$ = 34) and was –27.46‰ ± 0.26‰ ($n$ = 22) downstream of the mill. In the Kapuskasing River, δ¹³C upstream of the mill at KAP-WF was –31.02‰ ± 0.45‰ ($n$ = 8) and downstream of the mill at KAP-TB was –25.04‰ ± 0.25‰ ($n$ = 8).

### 2011 fish muscle and gonad δ¹³C

In 2011 there were differences in the muscle δ¹³C of white suckers captured downstream of the former SRF outfall in the Mattagami River compared with fish captured upstream. The analysis of tissue and site in 2011 found that muscle of white sucker captured at MAT-DS had significantly enriched ¹³C ($p < 0.001$; Fig. 2) compared with muscle of white sucker captured at MAT-US. In 2011 the δ¹³C of gonad tissue of white sucker was not significantly different between MAT-US and MAT-DS ($p = 0.067$). In the pooled group of males and females, there was significant enrichment of ¹³C in muscle compared with gonad tissue at MAT-DS ($p = 0.008$), but not at the reference site MAT-US ($p = 0.28$). These data suggest there was slower turnover of muscle compared with gonad tissue.

In the white sucker captured in the Kapuskasing River in 2011, a two-way ANOVA of site and tissue indicated a significant interaction between these factors ($p < 0.001$), suggesting that gonad and muscle responded differently among the sites. In the pooled group of males and females, muscle was significantly enriched in ¹³C compared with gonad in fish captured at KAP-TB ($p \leq 0.001$), but there

**Fig. 2.** Mean δ¹³C (‰) ± SE in muscle and gonad tissue of pooled female and male white sucker collected in the (*a*) Mattagami and (*b*) Kapuskasing rivers during the 1990s and 2011. Lights bars are reference sites (MAT-US or KAP-WF); dark bars are exposed sites (MAT-DS or KAP-TB). Within panes, shared uppercase letters denote no significant ($\alpha$ = 0.05) differences between tissues and sites in 2011; lowercase letters denote no significant differences in δ¹³C of muscle tissue between periods and sites. The horizontal shaded areas represent the mean of all fish captured at the reference sites in the respective river.




were no differences at the reference site KAP-WF ($p = 0.90$; Fig. 2). When compared between sites, δ¹³C of muscle and gonad was enriched at KAP-TB compared with KAP-WF ($p < 0.001$; Fig. 2). These data show that the patterns detected in δ¹³C in the 1990s downstream of the Kapuskasing mill are intact in 2011, but gonad tissue is more depleted than muscle tissue at KAP-TB. This pattern suggests a role of yearly turnover that may affect isotopic ratios in gonad tissue.

### δ¹³C: comparisons between periods in muscle tissue

There were significant differences between the open and closed periods at the SRF mill (Mattagami River) in the muscle tissue of white sucker (Fig. 2). In an ANOVA of period and site, there was a significant interaction between these factors on δ¹³C ($p < 0.001$), suggesting differences within sites in δ¹³C of muscle tissue between periods. In the muscle tissue of a pooled group of male and female white sucker captured at the MAT-DS site, ¹³C was significantly depleted in 2011 compared with muscle from fish collected during the 1990s at MAT-DS ($p < 0.001$), suggesting a change in their exposure to pulp mill effluent. At MAT-US there was no difference in the δ¹³C of muscle tissue in white sucker compared between the two sampling periods ($p = 0.18$), suggesting consistency of exposure at this site over time.

The Kapuskasing pulp mill has discharged effluent during both periods of study. A two-way ANOVA on δ¹³C indicated no significant interaction between site and period ($p = 0.63$), suggesting consistency of exposure between years. δ¹³C in muscle tissue was significantly elevated at KAP-TB compared with KAP-WF ($p < 0.001$) in both study periods (Fig. 2). Within each site sampled in the Kapuskasing River, δ¹³C in muscle tissue of white sucker was higher in the 1990s compared with that collected in 2011 ($p \leq 0.039$), but the differences were of small magnitude (1.34‰) and is within the

**Table 3.** Results of regressions of δ$^{13}$C (‰) in muscle and gonad and age (years) of white sucker captured at MAT-US, MAT-DS, KAP-WF, and KAP-TB in 2011.

| Tissue | Site | $r^2$ | $p$ | Slope | $y$ intercept |
|---|---|---|---|---|---|
| Muscle | MAT-US | 0.034 | 0.311 | −0.106 | −33.73 |
| | MAT-DS | 0.212 | 0.003 | 0.208A | −33.42A |
| | KAP-WF | 0.034 | 0.370 | 0.119 | −32.66 |
| | KAP-TB | 0.094 | 0.183 | 0.083b | −27.70a |
| Gonad | MAT-US | 0.126 | 0.054 | −0.110 | −32.95 |
| | MAT-DS | 0.126 | 0.039 | 0.210A | −34.70B |
| | KAP-WF | 0.156 | 0.059 | −0.188 | −31.33 |
| | KAP-TB | 0.225 | 0.019 | 0.239b | −32.12b |

**Note:** Uppercase letters denote differences between tissues at MAT-DS in slope and intercept, respectively; lowercase letters denote differences between tissues at KAP-TB in slope and intercept, respectively.

range of variation expected at a reference site over time (see Fig. 2). These data suggest that exposure regimes of fish in the Kapuskasing River between periods is similar.

### Relationship of δ$^{13}$C and age for muscle and gonad tissue

Regressions were conducted between δ$^{13}$C and age of fish captured in 2011. There was no difference in sex at MAT-US or MAT-DS ($p \geq 0.070$); sexes were pooled for further analyses. In this pooled group of males and females, the δ$^{13}$C of muscle and gonad tissues at MAT-DS were significantly related to age ($p \leq 0.039$; Table 3; Fig. 3), but there were no statistical relationships in either tissue at MAT-US ($p \geq 0.054$). These data suggest a difference in exposure patterns between young and old fish at MAT-DS that is not present at MAT-US.

There was no difference in δ$^{13}$C and age between sexes at the KAP-TB or KAP-WF sites ($p \geq 0.063$). The regressions of muscle δ$^{13}$C against age was not significant at the KAP-TB site ($p$ = 0.183; Table 3; Fig. 3), but the regression of gonad and age was ($p$ = 0.019). The regressions of gonad and muscle tissue were not significantly related to age at KAP-WF ($p \geq 0.059$).

ANCOVA was used to compare the slopes and intercepts between tissues at each of the two downstream sites: MAT-DS and KAP-TB. At MAT-DS, there was no difference in slope between δ$^{13}$C of muscle and δ$^{13}$C of gonad with age ($p$ = 0.94), but there was a significant difference in intercept ($p < 0.001$; Table 3). Similarly, at KAP-TB there was no difference in the slope of the regression lines of δ$^{13}$C and age between tissues ($p$ = 0.25), but there was a significant difference in the intercept ($p < 0.001$). This data suggests that there may be similar rates of tissue accrual between muscle and gonad, but that total turnover rates may be different because of some other activity, such as spawning.

### Opercular δ$^{13}$C

Using a pooled group of males and females, there was no relationship between year and opercular isotopes at MAT-US ($p$ = 0.852; $r^2 < 0.001$; Fig. 4). There was, however, a significant change in δ$^{13}$C compared with year in fish captured at MAT-DS ($p$ = 0.003; $r^2$ = 0.186), suggesting an effect of the closure of the mill on the C used during ossification. The δ$^{13}$C of opercular bone in white sucker captured at MAT-DS began drifting towards the value at MAT-US in 2007 and intersected the mean value of δ$^{13}$C at MAT-US in 2009.

## Discussion

The historical influence of C from effluents discharged into the Mattagami and Kapuskasing rivers was evident based on the $^{13}$C enrichment of white sucker muscle found downstream of both pulp mills in the 1990s. In 2011, pulp-derived C was still found in the muscle of white sucker captured downstream of the operational mill on the Kapuskasing River. While there was a change over time in the δ$^{13}$C of muscle of white sucker captured at the MAT-DS site, the

**Fig. 3.** Stable carbon (δ$^{13}$C, ‰) of muscle (*a*, *c*) and gonad (*b*, *d*) of white sucker collected in the Mattagami (*a*, *b*) and Kapuskasing (*c*, *d*) river sites compared with age; results of regressions are found in Table 3.

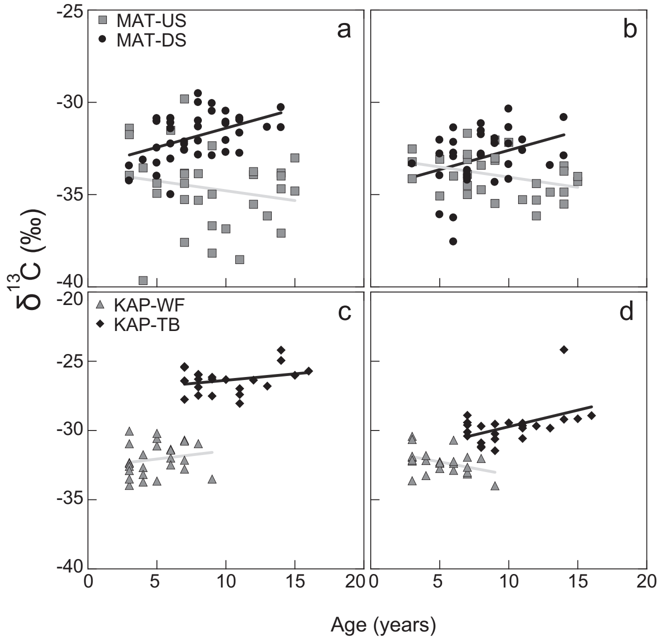
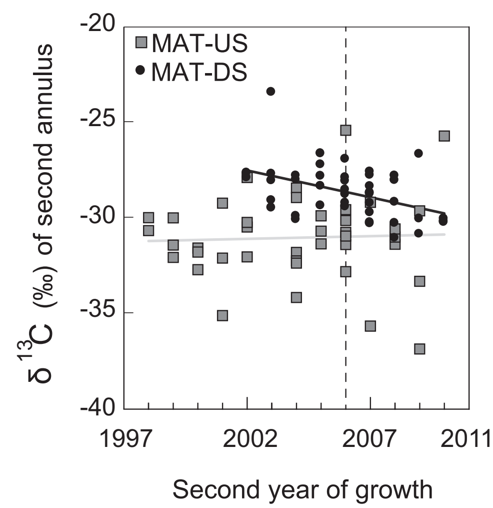



**Fig. 4.** δ$^{13}$C (‰) of second annulus compared against the second year of growth; the Mattagami mill was closed in 2006 (shown by the vertical dashed line).




disappearance of δ$^{13}$C associated with the pulp mill after its closure was occurring, but was incomplete. The residual δ$^{13}$C in the tissues of white sucker captured in the Mattagami River was related to age, illustrating the persistence of pulp-derived C in older suckers. This age-related pattern in the fish captured at the downstream site in the Mattagami River suggests the role of tissue amassed during the operation of the mill and its slower-than-expected turnover since closure. The occurrence of a residual signal after the removal of an enriching effluent from an environment may be a useful tool in the study of ecosystem recovery.

The use of stable isotopes to document recovery rates is affected by the biology of the organisms used to indicate the changes. The detection of an intermediate signal in δ$^{13}$C after the closure of the Mattagami mill is likely an effect of the incomplete turnover of tissues in sexually mature suckers since closure of the Mattagami mill. Fish captured at MAT-DS in 2011 that showed residual enrichment of muscle tissue tended to be at least 9 years old; these fish likely matured prior to the closure of the mill. In white sucker the somatic growth rate peaks at or around maturity (3–5 years of age; Munkittrick et al. 1991) and continues, albeit at a slowing rate, for the life of the fish (Ruemper 1998). This feature of slow turnover in older fish allowed the detection of residual C derived from the pulp mill in the muscle of white sucker. These data suggest the ongoing recovery of fish previously exposed to pulp mill effluent and the utility of $\delta^{13}C$ as a tool to document that recovery. The data also suggest that complete removal of the pulp-derived C from the receiving environment may require processes occurring over time, such as mortality or emigration of older fish.

Physiological processes are not, however, the only influence on $\delta^{13}C$ in the tissues of organisms. Ecological processes may affect the expected changes in $\delta^{13}C$ over time. The sampling of reference fish from a reservoir in the Mattagami River may influence the expected convergence of $\delta^{13}C$ among sites over time. The ratio of stable C in the tissues of organisms sampled from deeper waters tend to be more depleted compared with shallow areas, but this relationship is variable (Dauby 1989). These differences linked to depth are potentially associated with changes in photic environment (MacLeod and Barton 1998) and the presence of respired C (France 2000), which are both linked to depleted C signatures. This suggests that releasing water from reservoirs and the subsequent aeration may promote equilibration of the depleted aquatic $\delta^{13}C$ (Post 2002) with the enriched atmosphere (McConnaughey et al. 1997), which may enrich the former in the downstream aquatic environment. The degree to which this occurs is, however, not known. The overlapping signatures of $\delta^{13}C$ in young fish and its divergence in older fish suggests that the dam at SRF does not have an appreciable effect on the $\delta^{13}C$ of fish among those sites. Other larger dams in the area, however, appear to affect the $\delta^{13}C$ of fish captured in reservoirs and in the downstream areas (T. Arciszewski, unpublished data). This suggests that other factors, such as the depth of the reservoir, may be important in determining the possibility of similarity of $\delta^{13}C$ between reference sites in reservoirs and downstream riverine sites. Subsequently, this may affect the detection of recovery.

Comparing an upstream reference site with a downstream exposed site creates many issues in environmental monitoring, including an unknown effect on detecting real recovery at one specific and previously exposed site. In our case, changes at the exposed site, MAT-DS, are compared with the upstream reference site, MAT-US. Other studies have found changes in the ratios of stable isotopes after the closure of a facility (Tucker et al. 1999), but most, like ours, use a reference site to set a target for detecting recovery. A possible consideration in the detection of recovery using age as a covariate is the use of a target from within a site. Internal targets are the most relevant in a monitoring framework for detecting site-specific changes. In our work we found a statistical relationship between $\delta^{13}C$ of muscle and age only at the formerly exposed site, MAT-DS. This suggests that recovery can also be studied and detected without the use of a reference site. The presence of an age–tissue $\delta^{13}C$ relationship suggests that recovery is ongoing. The disappearance of that relationship over time would suggest that some aspects of recovery are complete. This relationship was also present in gonad $\delta^{13}C$ and age at MAT-DS, but we also observed a statistical relationship of gonad $\delta^{13}C$ and age at KAP-TB, suggesting that not all tissues can be used in this type of a comparison. The relationship observed at KAP-TB and the marginal significance at the reference sites between gonad $\delta^{13}C$ and age suggests complex changes in gonad tissue over time, questions of where that tissue is sourced, and the sensitivity to seasonal fluctuations in factors such as temperature and light intensity that can affect ratios of stable isotopes (MacLeod and Barton 1998) in metabolically active tissues. Although not useless, the interpretation of $\delta^{13}C$ data from gonad tissue is more complicated than that of $\delta^{13}C$ in muscle.

Another factor that may affect the persistence of the $\delta^{13}C$ from the Mattagami pulp mill in the fish captured in the downstream area is lag at the base of the food web. The opercular isotope data, which should closely reflect the $\delta^{13}C$ of a fish's diet in a given year, suggests that the $\delta^{13}C$ associated with fish captured downstream of the pulp mill in the 1990s was present in the food of white sucker beyond the closure date of the mill. Organic matter derived from the pulp mill may still be present in the sediments at the MAT-DS site (i.e., Costa et al. 1979), but our techniques may not be sensitive enough to detect its influence. Concurrent work on invertebrates suggests that some eaten by white sucker downstream of the mill after its closure have a residual influence, but these patterns are clearer in $\delta^{15}N$, not $\delta^{13}C$ (H. McMahon, unpublished data).

Opercular data suggested that the influence of pulp-derived C disappeared by 2009, but there may also be a possible role of scavenging of existing tissues during ossification and bone remodelling in the years immediately after the closure of the mill. The disappearance by 2009 suggests the primary role of new materials, derived from either the diet or existing tissues with a depleted $\delta^{13}C$, in ossification that year. Directly comparing $\delta^{13}C$ in bone with other tissues or what is present in the environment may not be without issues. $\delta^{13}C$ of the second annulus of each fish captured at either MAT-DS or MAT-US was slightly higher compared with the $\delta^{13}C$ from fish muscle measured during the 1990s. Enrichment of $^{13}C$ in bone compared with muscle was also observed in a previous study in the Moose River basin (Farwell 2000) and elsewhere (Dubé et al. 2005). This enrichment suggests the possible routing and accumulation of particular isotopes to different compartments in the fish's body via specific compounds or the uptake and incorporation of $^{13}C$-enriched inorganic C (Solomon et al. 2008). The amino acids required for the type of collagen in fish bone may be more $^{13}C$-enriched compared with muscular or gonadal proteins (Wolf et al. 2009) or the required proteins for either compartment are limited in the diet (Wilson 2002). However, little is known about the metabolism of bone in fish (i.e., Lall and Lewis-McCrea 2007).

We expected gonad tissue of fish captured in the Mattagami River to equilibrate with the historical reference value of $\delta^{13}C$ quickly. In support of this hypothesis, the mean $\delta^{13}C$ of gonad tissue in white sucker captured at this site after mill closure was within the historical reference range calculated from fish collected at the MAT-US site. There was, however, also an age-related pattern in the gonad tissue of white sucker captured at the MAT-DS site; when accounting for age, however, the older fish still show residual $\delta^{13}C$. These conclusions are not necessarily mutually exclusive. Because the relationship of age and gonad $\delta^{13}C$ was weaker than age and muscle $\delta^{13}C$, the similarity we found in gonad tissue of white sucker in $\delta^{13}C$ between sites in the Mattagami River when not accounting for age was likely affected by the seasonal development of gonad tissue. Spawning is, functionally, a mechanism of turnover in gonad tissue, but not all of the accumulated eggs or sperm may be shed each year. The residual C in gonad tissues of the older white sucker may be the influence of structural, somatic, and persistent, rather than gametic and transient, components of testes and ovaries (Yaron and Sivan 2006). Whether the proportion of somatic tissue in the gonad increases with age in fish is not clear, but would also explain the pattern we observed. A further explanation may be changes in feeding rates with age and a higher reliance on scavenging existing tissues for gonadal development in older fish compared with younger fish.

The lag in $\delta^{13}C$ after the closure of the mill could be a useful monitoring tool. The utility partly stems from persistence of old $\delta^{13}C$ signatures in hard or soft tissues of organisms that can overcome the lack of "before" data. Although it has been advocated repeatedly (i.e., Lindenmayer and Likens 2010), few monitoring programs have consistent and continuous data from before development begins through operation and closure. To overcome these deficiencies, hard tissues have been suggested as useful archives of information obtained from stable isotope ratios (Grønkjær et al. 2013). This natural archiving can accommodate the construction of historical ranges of $\delta^{13}C$ signatures in natural environments. We did, however, find age-related patterns in soft tissues that mimic a temporal gradient. Between the soft tissues, most compelling for the detection of residual patterns is the age-specific relationship of changes in $\delta^{13}C$ away from the historical signature associated with the closed mill. Although we had pre-closure data available, in muscle we observed a pseudo-temporal gradient within the fish collected in 2011 using a broad age range of individuals. This pseudo-temporal gradient depends on a predictable decline of tissue turnover rate with age after maturity. We were able to demonstrate that tissue libraries are not necessary to document recovery or ecosystem changes linked to an event. Longer-lived organisms like fish can be used as living archives to indicate long-term disturbance, especially in fish that spawn each year and may show a divergence in a focal characteristic among different cohorts of fish, such as gonad size. For instance, our data shows that fish younger than 5 years have overlapping $\delta^{13}C$ among sites in 2011. These fish likely matured after the closure of the mill. Correspondingly, the data also shows that fish older than 5 years (and likely matured before 2006) are more likely to show residual C associated with exposure to the pulp mill's effluent, and this likelihood increases with age. Differences in $\delta^{13}C$ among ages of fish may mirror differences among organisms with varying rates of individual turnover and can likely be exploited as a monitoring tool.

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